



High Voltage Stability of Interfacial Reaction at the LiMn_2O_4 Thin-Film Electrodes/Liquid Electrolytes with Boroxine Compounds

T. Horino, H. Tamada, A. Kishimoto, J. Kaneko, Y. Iriyama,^{*,z}
Y. Tanaka, and T. Fujinami^{*}

Department of Materials Science and Chemical Engineering, Faculty of Engineering, Shizuoka University,
Hamamatsu, Shizuoka 432-8561, Japan

Liquid electrolytes with boroxine compounds were prepared, and interfacial reactions at the liquid electrolytes/ LiMn_2O_4 thin-film electrodes were investigated. Boroxine compounds, $\text{B}_3\text{O}_3(\text{OR})_3$, with different kinds of substituents [$\text{R} = \text{CH}_3$, CH_2CH_3 , $(\text{CH}_2)_2\text{CH}_3$, and $\text{CH}(\text{CH}_3)_2$] were dissolved in 1 mol kg^{-1} LiCF_3SO_3 /ethylene carbonate + EMC (1:1 by volume). Both cyclic voltammetry and electrical impedance spectroscopy measurements revealed that interfacial resistance depended on the chain length in the substituent of the boroxine compounds, and the resistance decreased with the increase in the chain length. Moreover, the anodic stability of the electrolytes was improved up to 5.0 V (vs Li/Li^+) on the film electrodes by the addition of tri-isopropoxy boroxine [$\text{R} = \text{CH}(\text{CH}_3)_2$] with a steric structure (branched chain substituent). These results indicate that mixing boroxine compounds with both appropriate chain length and steric structure can be an effective way to develop advanced liquid electrolytes for high voltage rechargeable lithium ion batteries.

© 2010 The Electrochemical Society. [DOI: 10.1149/1.3374638] All rights reserved.

Manuscript submitted November 9, 2009; revised manuscript received March 4, 2010. Published April 21, 2010.

Rechargeable lithium ion batteries have been expected as power sources for hybrid and electric vehicles, and some of them have been commercialized. However, further improvements in the current battery system are strongly demanded at the social end, and one of the most expected properties is the enhancement of energy density to prolong the cruising distance of electric vehicles. In this prospect, increase in battery operating voltage must be a possible way because there are several kinds of 5 V class positive electrode materials such as $\text{LiNi}_{0.5}\text{Mn}_{1.5}\text{O}_4$,¹ LiCoPO_4 ,² and LiCoMnO_4 .³

One of the drawbacks to realize 5 V class rechargeable lithium ion batteries is to develop advanced electrolytes with a wide potential window, high ionic conductivity, sufficient safety, low cost, etc. Although new kinds of electrolytes, typically ionic liquids⁴ and solid electrolytes,⁵ have been proposed, several problems such as cost, power density, and large resistance at the interface still remained to put them in practical use.

In addition to the developments of the above electrolytes, various boron-based compounds have been studied as a new class of electrolytes. Zhang and Angell discovered the electrolytes with boron-based cosolvent. They showed that the electrolyte obtains an approximately 5 V stability on platinum electrodes and that the electrolyte can work well in both Li/LiMnO_2 and carbon/ LiMnO_2 cells at 2.0–4.4 V.⁶ Barthel et al. proposed new lithium salts with a chelate-type anion of boron.^{7,8} Sasaki et al. prepared their derivatives and applied them to a practical lithium ion battery system.^{9,10} Although the anodic stability of these lithium salts was ca. 4.2 V vs Li/Li^+ , Xu and Angell discovered lithium bis(oxalate) borate (LiBOB), which obtained superior anodic stability up to 4.5 V vs Li/Li^+ on platinum electrode.¹¹ The LiBOB obtained other interactive properties for lithium ion batteries such as high stability of LiMn_2O_4 at high temperature,¹² lower corrosive properties to Al current collector,¹³ and formation of stable solid electrolyte interphase (SEI) on graphite negative electrode.¹⁴ Recently, Li et al. proposed new electrolytes where boron-based anion receptors interact with LiF , which obtains an anodic stability of up to 5.0 V vs Li/Li^+ on a platinum electrode.¹⁵

Although boron-based compounds have possibilities to develop advanced liquid electrolytes as mentioned above, boroxine compounds that can be easily prepared with low cost have not been systematically investigated at all especially in liquid electrolytes. Boroxine compound obtains Lewis acidity, and its interaction with

lithium salt can promote the dissociation reaction of the salt and also increase the lithium ion transference number in its conductivity.¹⁶ Mehta and Fujinami applied this interaction to prepare polymer electrolytes with high ionic conductivities and high lithium transference number.^{16–18} Here, we developed liquid electrolytes dissolving different kinds of boroxine compounds and investigated the interfacial reaction at the liquid electrolytes/ LiMn_2O_4 thin films especially focusing on the high voltage stability. We showed that the anodic stability of the liquid electrolyte [1 mol kg^{-1} LiCF_3SO_3 /ethylene carbonate (EC) + EMC (1:1 by volume)] can improve up to 5.0 V on a LiMn_2O_4 thin-film electrode by dissolving tri-isopropoxy boroxine in it.

Experimental

The molecular structure of the boroxine compounds [$\text{B}_3\text{O}_3(\text{OR})_3$] studied in this work is shown in Fig. 1. These boroxine compounds, trimethoxy boroxine ($\text{R} = \text{CH}_3$), triethoxy boroxine ($\text{R} = \text{CH}_2\text{CH}_3$), trinormalpropoxy boroxine [$\text{R} = (\text{CH}_2)_2\text{CH}_3$], and tri-isopropoxy boroxine [$\text{R} = \text{CH}(\text{CH}_3)_2$], are denoted as TMBx , TEBx , TnPBx , and TiPBx , respectively, in subsequent descriptions. The substituent (R) of the former three compounds is a straight chain, while the last one, TiPBx , a branched chain.

The TMBx was a reagent grade chemical (Aldrich, 98%), but the other boroxine compounds were prepared in laboratory scale by mixing boron trioxide (Aldrich) with different kinds of alkyl borate (Tokyo Kashei). All the alkyl borates were well-distilled before use. The resultant boroxine compounds were dissolved in 1 mol kg^{-1} LiCF_3SO_3 /EC + EMC (1:1 by volume) in an argon-filled glove box so that the concentration of boroxine compounds in the electrolytes became the same with that of LiCF_3SO_3 . Hereafter, the liquid electrolyte without boroxine compounds and that with TMBx , TEBx , TnPBx , and TiPBx are denoted as $\text{EL}(\text{original})$, $\text{EL}(\text{TMBx})$, $\text{EL}(\text{TEBx})$, $\text{EL}(\text{TnPBx})$, and $\text{EL}(\text{TiPBx})$, respectively. Ionic conductivities of the liquid electrolytes were measured by ac impedance spectroscopy using platinum blocking electrodes. Lithium ion transference numbers were measured by sandwiching the electrolytes between lithium metal sheets using the combined ac impedance/dc polarization method of Evans et al.¹⁹ modified by Abraham et al.²⁰

Thin films of LiMn_2O_4 (ca. 60 nm thick) were prepared by pulsed laser deposition. The films were deposited for 2 h on Pt substrates heated at 973 K, and the resultant films were characterized by X-ray diffraction (XRD).

The electrochemical reaction of the film electrodes in the above electrolytes was investigated by a three-electrode cell where the working electrodes were the LiMn_2O_4 thin-film electrodes (area of

^{*} Electrochemical Society Active Member.

^z E-mail: tyiriy@ipc.shizuoka.ac.jp

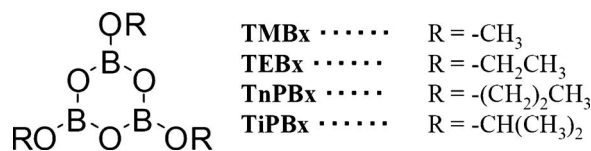


Figure 1. Schematic molecular structure of boroxine compounds used in this study.

0.50 cm²), and both the counter and reference electrodes were lithium metal. Hereafter, all the potentials are referred to Li/Li⁺. Cyclic voltammetry (CV) was carried out between 3.5 and 4.5 V or between 3.5 and 5.0 V at a potential sweep rate of 0.5 mV s⁻¹. Electrochemical impedance spectroscopy (EIS) was conducted at 3.95 V after the electrode potential was kept at the potential for 10 min to attain the electrode at a steady state. All the EIS measurements were conducted by applying a sine wave with an amplitude of 10 mV (root-mean-square) over the frequency range of 200 kHz to 10 mHz. To evaluate the stability of interfacial resistance at 5.0 V, the electrode potential was swept to 5.0 V at a rate of 0.5 mV s⁻¹ and was kept at the potential for a given time. After that, the potential was swept again to 3.95 V at the same potential sweep rate, and the EIS data were collected in the same manner. Also, the electrochemical lithium plating–stripping reaction on the Ni plate was conducted in some electrolytes to investigate their cathodic properties. Lithium was plated on the Ni plate for 5,000 s by 5 μA cm⁻², following the stripping reaction by the inverse current density until the electrode potential reached 3 V. This lithium plating–stripping reaction was repeated for 50 times. EIS was measured at the initial voltage and at the charged states during the plating–stripping reactions. All these electrochemical measurements were conducted in an Ar-filled glove box at room temperature.

Results and Discussion

Basic properties of liquid electrolytes and thin-film electrodes.—

Figure 2 shows the variation in the ionic conductivities of the electrolytes with different kinds of boroxine compounds. The conductivities measured over 273 K decreased by the addition of boroxine compounds, while the conductivities under 253 K increased basi-

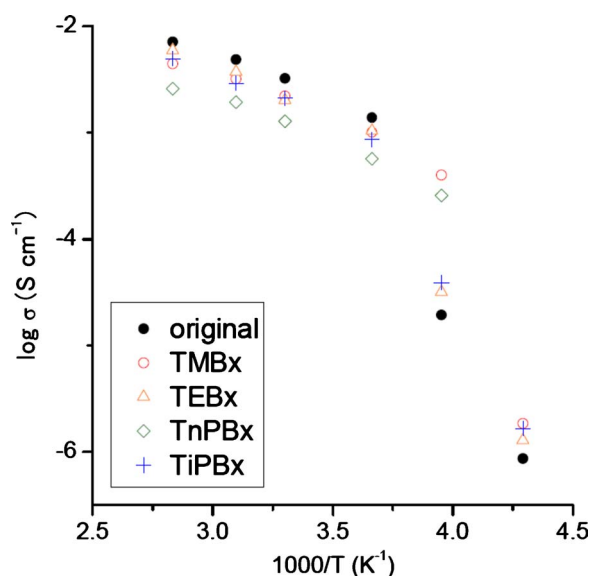


Figure 2. (Color online) Temperature dependency of ionic conductivities of 1 mol kg⁻¹ LiCF₃SO₃ dissolved in EC + EMC (1:1 by volume) with equivalent concentration boroxine compound; original electrolyte without boroxine compounds (●) and the electrolytes with TMBx (○), TEBx (Δ), TnPBx (◇), and TiPBx (+).

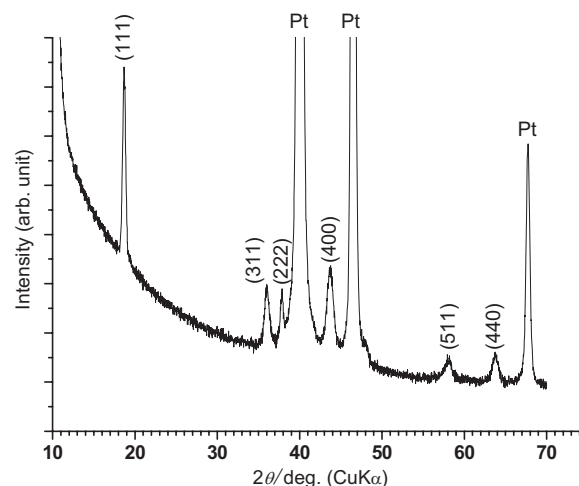


Figure 3. XRD pattern of LiMn₂O₄ thin film deposited on platinum substrate.

cally. The resistance of EL(TnPBx) at 233 K was too large to measure probably because of the crystallization of the electrolyte. Boroxine ring obtains Lewis acidity, and the ring structure can strongly interact with CF₃SO₃⁻ anion.¹⁶ Although this interaction was usefully applied to develop a polymer electrolyte with high ionic conductivity and high lithium ion transference number,^{16–18} mixing boroxine compounds in those liquid electrolytes increased their viscosities (η) ($\eta_{303\text{ K,EL(original)}} = 3.5\text{ mPa s}$, $\eta_{303\text{ K,EL(TMBx)}} = 4.6\text{ mPa s}$, $\eta_{303\text{ K,EL(TEBx)}} = 5.3\text{ mPa s}$, $\eta_{303\text{ K,EL(TnPBx)}} = 5.3\text{ mPa s}$, and $\eta_{303\text{ K,EL(TiPBx)}} = 5.4\text{ mPa s}$ at room temperature). However, lithium ion transference number increased over 0.5 by mixing boroxine compounds, which is larger than those of general organic liquid electrolytes (ca. 0.3).¹⁵ Both the increase in viscosity and the decrease in anion transference number would contribute to the variation in conductivities.

Figure 3 shows the XRD patterns of the LiMn₂O₄ thin film on the Pt substrate. The diffraction peaks, except for that of the substrate, were all assigned to spinel-structured LiMn₂O₄, as indexed in Fig. 3, and any impurity peaks were not observed. This result agreed well with a previous paper.²¹

Effects of chain structure in the substituent on the interfacial resistance at the LiMn₂O₄ thin-film electrodes/liquid electrolytes with boroxine compounds.—Figure 4a summarizes the CVs of LiMn₂O₄ thin-film electrodes at 3.5–4.5 V in different kinds of liquid electrolytes. Two couples of redox peaks assigned to the lithium insertion/extraction reaction into LiMn₂O₄ were clearly observed in every case. The peak potential difference (ΔE_{ac}) of each redox reaction increased by mixing boroxine compounds and showed the dependency of their substituent species. In the lower potential redox peak, the ΔE_{ac} was 22 mV in the EL(original), and in the electrolytes with boroxine compounds, the ΔE_{ac} decreased with the increase in chain length in the substituent [ΔE_{ac} in the EL(TMBx): 138 mV, ΔE_{ac} in the EL(TEBx): 109 mV, ΔE_{ac} in the EL(TnPBx): 86 mV, and ΔE_{ac} in the EL(TiPBx): 85 mV].

The EIS of LiMn₂O₄ thin films were measured to evaluate the interfacial resistance at 3.95 V, and typical Cole–Cole plots are shown in Fig. 4b. The plots consisted of potential-independent one semicircular arc at the high frequency region assigned to electrolyte resistance, potential-dependent one semicircular arc at the middle frequency region assigned to interfacial resistance, and nearly vertical line to the real axis due to the capacitive element. The interfacial resistance was clearly observed at 3.95 V while the resistance was not at 3.50 V. This is because the thin-film electrode does not show lithium insertion/extraction reaction so much at 3.50 V, as shown in

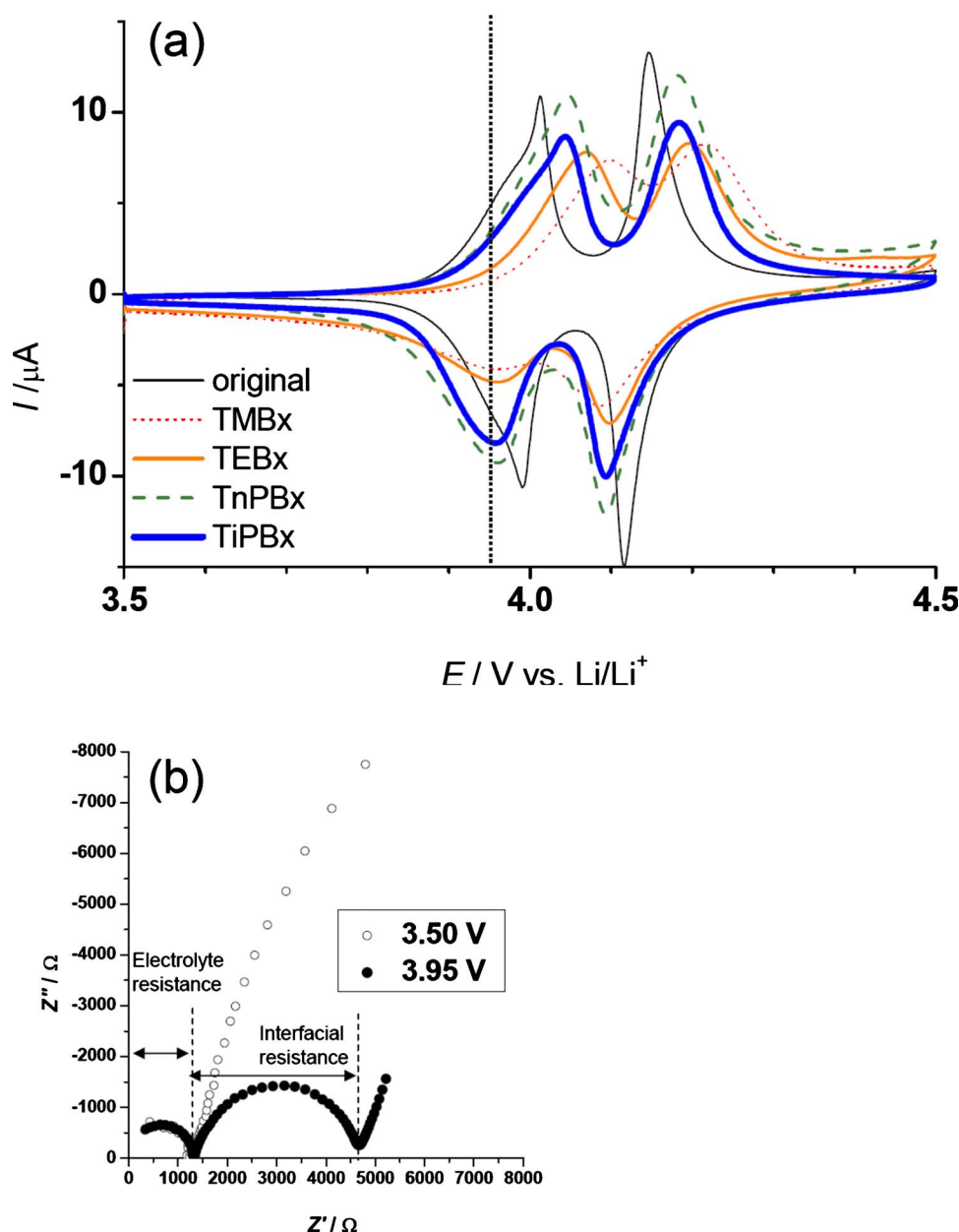


Figure 4. (Color online) (a) CVs (3.5–4.5 V) of LiMn_2O_4 at room temperature in $1 \text{ mol kg}^{-1} \text{ LiCF}_3\text{SO}_3$ dissolved in EC + EMC (1:1 by volume) without boroxine compounds (thin line) and the electrolyte with TMBx (thin dashed line), TEBx (medium line), TnPBx (medium broken line), and TiPBx (bold line) at $v = 0.5 \text{ mV s}^{-1}$. (b) Typical Cole–Cole plots obtained at 3.50 V (○) and at 3.95 V (●) in the electrolyte with and without boroxine compounds. These data were taken in the electrolyte with TnPBx.

Fig. 4a. Table I summarizes the interfacial resistances at 3.95 V in these electrolytes after CV measurement at 3.5–4.5 V for one cycle. Interfacial resistance in the electrolytes with boroxine compounds obtained a larger value than that in the EL(original). The interfacial resistance decreased with the increase in the chain length in the

substituent of boroxine compounds. Although these electrolytes have different conductivities at 303 K, as shown in Fig. 2, the electrolyte resistances varied from 1400 to 3000 Ω in assembled electrochemical cells, which were much smaller than the differences in interfacial resistance. Hence, it is concluded that the difference in the above ΔE_{ac} mainly originates from the difference in interfacial resistances, and chain length in the substituent of boroxine compounds is an important factor to decrease the interfacial resistance at LiMn_2O_4 thin-film electrodes/liquid electrolytes with boroxine compounds.

Electrochemical stability of the liquid electrolytes on LiMn_2O_4 thin-film electrode at 5.0 V.— Figure 5 shows the CVs measured at 3.5–5.0 V. Irreversible anodic currents were clearly observed at 5.0 V except for the electrolyte with TiPBx. In those electrolytes, interfacial resistance at 3.95 V also largely increased after the CV measurements at 3.5–5.0 V for one cycle, and cathodic current peaks decreased. However, in EL(TiPBx), such increase in interfacial resistance was remarkably suppressed, as shown in Table I, and cathodic current peaks of LiMn_2O_4 were observed as with the anodic

Table I. Variation in interfacial resistance (R_{ct}) of the LiMn_2O_4 thin-film electrodes at 3.95 V in $1 \text{ mol kg}^{-1} \text{ LiCF}_3\text{SO}_3$ dissolved in EC + EMC (1:1 by volume) with and without boroxine compounds. All the data were measured after the CVs at 3.5–4.5 V or at 3.5–5.0 V with a potential sweep rate of 0.5 mV s^{-1} for one cycle.

Cutoff voltage (V)	R_{ct} (Ω)				
	Original	TMBx	TEBx	TnPBx	TiPBx
4.5	396	24,015	10,683	4829	3122
5.0	1047	53,800	359,700	248,410	3827

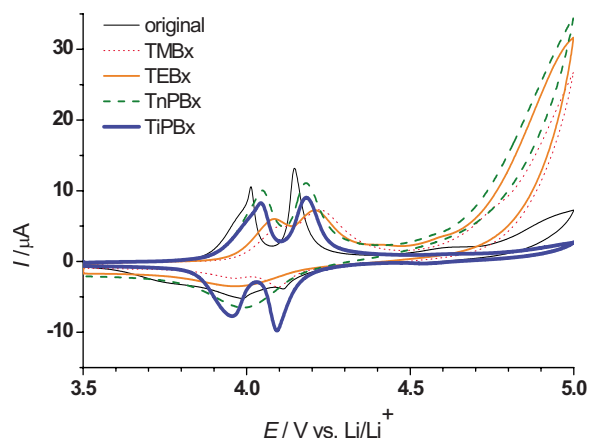


Figure 5. (Color online) CVs (3.5–5.0 V) of LiMn_2O_4 thin films at room temperature in $1 \text{ mol kg}^{-1} \text{ LiCF}_3\text{SO}_3$ dissolved in EC + EMC (1:1 by volume) without boroxine compounds (thin line) and the electrolyte with TMBx (thin dashed line), TEBx (medium line), TnPBx (medium broken line), and TiPBx (bold line). The data were measured after the CVs at 3.5–4.5 V for one cycle at $\nu = 0.5 \text{ mV s}^{-1}$.

ones. These results indicate that the stable lithium insertion/extraction reaction of LiMn_2O_4 can proceed in the EL(TiPBx) even when the electrode potential attain 5.0 V at the charging reaction.

To evaluate the above electrochemical stability more clearly, variation in interfacial resistance at 3.95 V against holding time at 5.0 V was measured, as shown in Fig. 6. The interfacial resistance clearly gradually increased with the increase in the holding time in the EL(original), while the variation was suppressed well in the EL(TiPBx).

As shown in Fig. 5, a larger amount of oxidation current flows in the electrolytes with boroxine compounds except for TiPBx compared to that in the EL(original). When the boroxine compounds interact with the CF_3SO_3^- anion, the electron density in the anion decreases, which must increase the anodic stability of anion itself. Therefore, it is reasonable to expect that the irreversible oxidation reactions observed at 5.0 V originate not from the enhancement of anodic decomposition reaction of anion itself but from the anodic irreversible decomposition reaction of boroxine compounds themselves. Here, alkyl chains in the substituent of boroxine compounds are generally considered as an electron-releasing group, and the longer chain substituent can easily release a larger amount of elec-

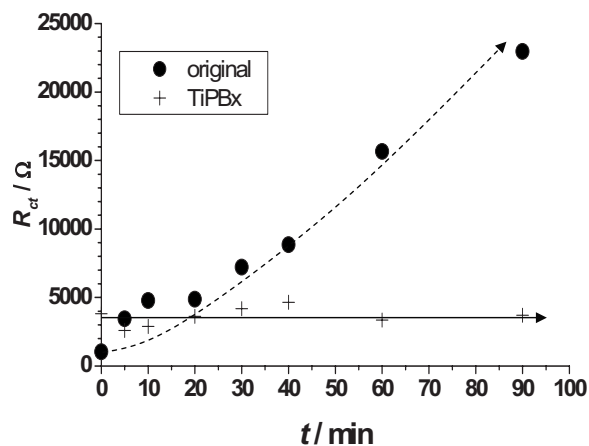


Figure 6. Variations in interfacial resistances at 3.95 V by the change in holding time at 5.0 V in $1 \text{ mol kg}^{-1} \text{ LiCF}_3\text{SO}_3$ dissolved in EC + EMC (1:1 by volume) without boroxine compounds (●) and in the electrolyte with TiPBx (○).

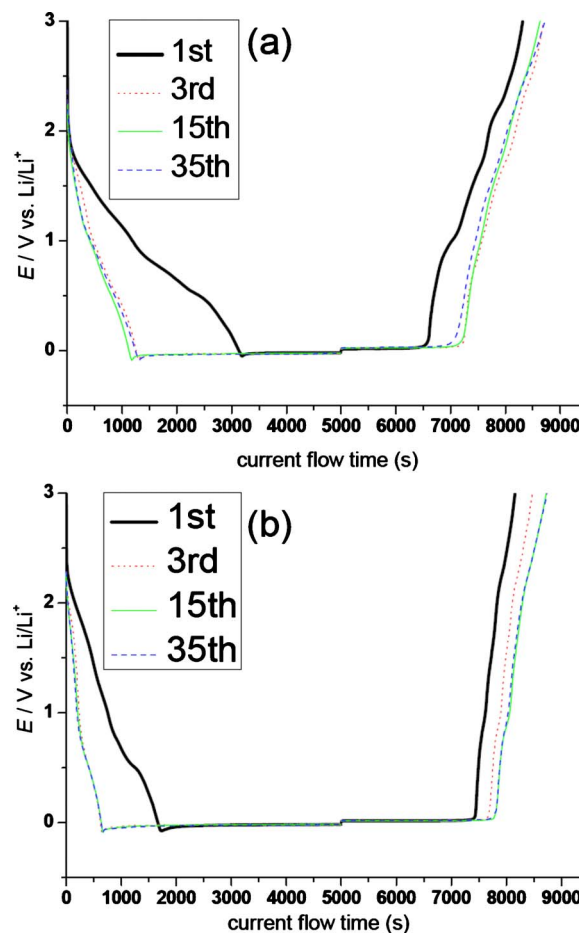


Figure 7. (Color online) Voltage curves of lithium plating and stripping reactions on Ni plates at room temperature in $1 \text{ mol kg}^{-1} \text{ LiCF}_3\text{SO}_3$ dissolved in EC + EMC (1:1 by volume) (a) without boroxine compounds (thin line) and (b) with TiPBx. The 1st (bold line), 3rd (thin dotted line), 15th (thin solid line), and 35th (thin broken line) reactions at $I = 5 \text{ μA cm}^{-2}$.

tron. Thus, the TMBx with the shortest chain length can be expected as the most stable material for the anodic reaction in those boroxine compounds. In fact, the anodic current in the EL(TMBx) at 5.0 V became the smallest value among the ELs with boroxine compounds except for the EL(TiPBx). This stability seems to be reflected also on the stability of interfacial resistance, as shown in Table I. Based on the above consideration, it is recognized that TiPBx with the longest chain length obtains quite unique properties in the boroxine compounds, and steric effect in the substituent that means branched chain substituent must play an important role for improving the high voltage stability.

Cathodic properties of the electrolyte with tri-isopropoxy boroxine.— In the previous section, the EL(TiPBx) obtains unique high voltage stability on the LiMn_2O_4 thin-film electrode. To evaluate the cathodic properties of the EL(TiPBx), electrochemical lithium plating–stripping reactions on Ni plates were measured by galvanostatic charge–discharge reaction for 50 times. Figure 7 shows the voltage curves of lithium plating–stripping on Ni plates at the 1st, 3rd, 15th, and 35th reactions in both the EL(original) and EL(TiPBx). In the beginning of the first charge (lithium plating reaction) curve, a gradual potential drop was observed in both electrolytes. The corresponding potential change was not observed in the later period of the first discharge (lithium stripping reaction) curve, and such unsymmetrical potential change disappeared after a few reactions. This is mainly because irreversible reductive decomposition reactions of electrolytes occur in this region, resulting in the

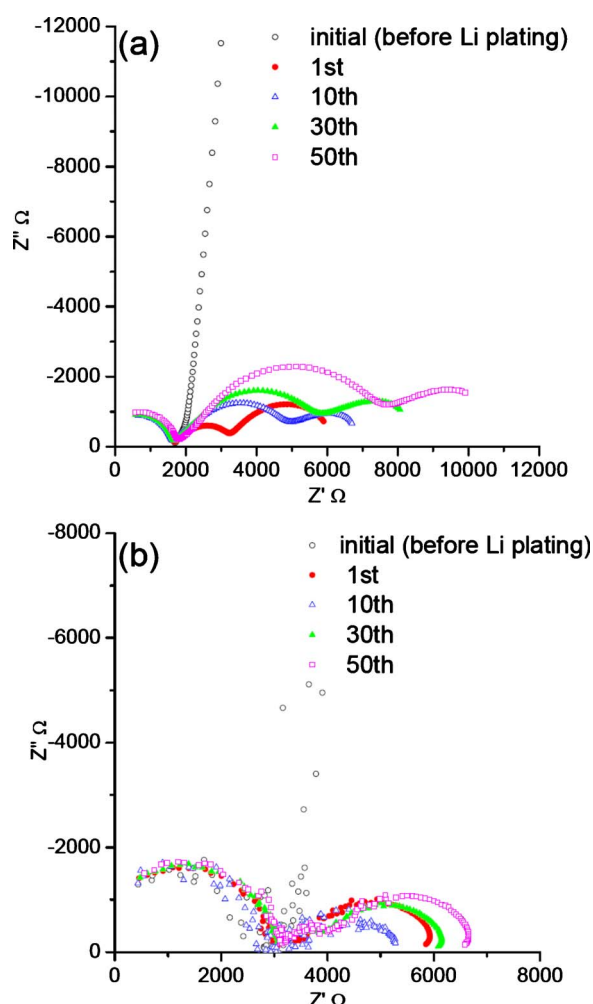


Figure 8. (Color online) Cole-Cole plots for lithium plating-stripping reaction in 1 mol kg⁻¹ LiCF₃SO₃ dissolved in EC + EMC (1:1 by volume) (a) without boroxine compounds (thin line) and (b) with TiPBx at the initial state (before Li plating (○) and after the 1st (●), 10th (△), 30th (▲), and 5th (□) charged reactions for 30 min at $I = 5 \mu\text{A cm}^{-2}$.

formation of SEI. Also, irreversible reduction of NiO on the surface of metal Ni is partially included in the irreversible capacity. Reversible capacities at potentials higher than that of Li plating-stripping observed after the third cycle are mainly due to the double-layer capacitance of the rough Ni surface. By removing the effect of the double-layer capacitance, the coulomb efficiency was estimated to be 55–60% in the EL(original), while the value slightly increased to 62–65% in the EL(TiPBx).

Figure 8 shows the Cole-Cole plots at the initial potential (approximately 3 V, before the lithium plating reaction) and after charged reactions for 30 min at the 1st, 10th, 30th, and 50th charge-discharge reactions shown in Fig. 7. Each spectrum consisted of mainly three semicircular arcs. The initial semicircular arc observed at the highest frequency region in both systems is assigned to the electrolyte resistance because this semicircular arc can be observed in the same manner before the lithium plating. The other two semicircular arcs observed at the lower frequency region are related to the lithium plating-stripping reaction process, such as surface film and charge-transfer reaction.^{22,23} As shown in Fig. 8, the diameter of the two semicircular arcs increased by the repetition of the charge-

discharge reaction in the EL(original) (Fig. 8a), but the resistance in the EL(TiPBx) became smaller and the increase in resistance with the reaction was suppressed (Fig. 8b). These results indicate that the EL(TiPBx) can stabilize the lithium plating-stripping reaction. Although same lithium plating-stripping reactions were conducted also in the electrolytes with other boroxine compounds, the irreversible capacity in the first charge-discharge reaction was very large. This may indicate another merit of TiPBx for cathodic properties, which are currently investigated.

Conclusion

Boroxine compounds with different chain lengths and steric structures in the substituent were dissolved in 1 mol kg⁻¹ LiCF₃SO₃/EC + EMC liquid electrolytes, and the interfacial reaction of the electrolytes with LiMn₂O₄ thin films was investigated. Interfacial resistance depended on the chain length in the substituent of boroxine compounds, and the resistance decreased with the increase in the chain length. Moreover, both the CV and EIS data revealed that the electrolyte-dissolving tri-isopropoxy boroxine (TiPBx) with a branched chain substituent suppressed the anodic irreversible current flow at 5.0 V (vs Li/Li⁺) and stabilized the interfacial resistance on LiMn₂O₄ thin films even at 5.0 V. The electrolytes with TiPBx also showed a stabilized lithium plating-stripping reaction at the cathode side. As with other boron-based materials, these results suggest that boroxine compounds with appropriate steric structure and chain length can be useful materials to develop advanced liquid electrolytes for high voltage-type rechargeable lithium ion batteries.

Acknowledgment

This work was supported by the New Energy and Industrial Technology Development Organization (NEDO) of Japan.

Shizuoka University assisted in meeting the publication costs of this article.

References

- Q. M. Zhong, A. Bonakdarpour, M. J. Zhang, Y. Gao, and J. R. Dahn, *J. Electrochem. Soc.*, **144**, 205 (1997).
- K. Amine, H. Yasuda, and M. Yamachi, *Electrochem. Solid-State Lett.*, **3**, 178 (2000).
- H. Kawai, M. Nagata, H. Kageyama, H. Tukamoto, and A. R. West, *Electrochim. Acta*, **45**, 315 (1999).
- H. Sakaebe and H. Matsumoto, *Electrochem. Commun.*, **5**, 594 (2003).
- R. Murugan, V. Thangadurai, and W. Weppner, *Angew. Chem., Int. Ed.*, **46**, 7778 (2007).
- S. S. Zhang and C. A. Angell, *J. Electrochem. Soc.*, **143**, 4047 (1996).
- J. Barthel, R. Buestrich, H. J. Gores, M. Schmidt, and M. Wuhr, *J. Electrochem. Soc.*, **144**, 3866 (1997).
- J. Barthel, M. Schmidt, and H. J. Gores, *J. Electrochem. Soc.*, **145**, L17 (1998).
- Y. Sasaki, S. Sekiya, M. Handa, and K. Usami, *J. Power Sources*, **79**, 91 (1999).
- Y. Sasaki, M. Handa, K. Kurashima, T. Tonuma, and K. Usami, *J. Electrochem. Soc.*, **148**, A999 (2001).
- W. Xu and C. A. Angell, *Electrochem. Solid-State Lett.*, **4**, E1 (2001).
- K. Amine, J. Liu, S. Kang, I. Belharouak, Y. Hyung, D. Vissers, and G. Henriksen, *J. Power Sources*, **129**, 14 (2004).
- H. Xie, Z. Tang, Z. Li, Y. He, H. Wang, and Y. Liu, *Electrochem. Solid-State Lett.*, **11**, C19 (2008).
- K. Xu, S. S. Zhang, B. A. Poesche, and T. R. Jow, *Electrochem. Solid-State Lett.*, **5**, A26 (2002).
- L. F. Li, H. S. Lee, H. Li, X. Q. Yang, K. W. Yang, W. S. Yoon, J. McBreen, and X. J. Huang, *J. Power Sources*, **184**, 517 (2008).
- M. A. Mehta and T. Fujinami, *Chem. Lett.*, **26**, 915 (1997).
- M. A. Mehta, T. Fujinami, and T. Inoue, *J. Power Sources*, **81–82**, 724 (1999).
- M. A. Mehta, T. Fujinami, S. Inoue, K. Matsushita, T. Miwa, and T. Inoue, *Electrochim. Acta*, **45**, 11175 (2000).
- J. Evans, C. A. Vincent, and P. G. Bruce, *Polymer*, **28**, 2324 (1987).
- K. M. Abraham, Z. Jiang, and B. Carroll, *Chem. Mater.*, **9**, 1978 (1997).
- I. Yamada, T. Abe, Y. Iriyama, and Z. Ogumi, *Electrochem. Commun.*, **5**, 502 (2003).
- D. Aurbach and A. Zaban, *J. Electrochem. Soc.*, **141**, 1808 (1994).
- K. Kanamura, S. Shiraishi, and Z. Takehara, *J. Electrochem. Soc.*, **143**, 2187 (1996).